Cover Letter

To: Editorial Office, *Quantum Science and Technology* (IOP Publishing) **From:** Aleaddin Özer (Corresponding Author) · ORCID 0000-0001-9389-5357 **E-mail:** ozer@e2esolutions.tech **Date:** 1 June 2026

Dear Editors,

We are submitting *"Three Axes of Quantum Risk: A Unified Observability Model for PQC, QKD, and Crypto-Agility"* for consideration in *Quantum Science and Technology*.

PQ-readiness scanners today answer one question: is the primitive quantum-resistant? Two assets that look identical under that question can have very different migration costs, and very different channel-level protections, neither of which the scanner sees. The paper treats quantum-risk posture as a three-axis problem instead, scoring each asset on algorithmic resistance, channel protection, and migration agility, then folding the three into a deadline-adjusted risk score q. We also release the reference implementation end to end: eBPF-based TLS observation, a working ETSI GS QKD 014 collector with a KME emulator, and a static crypto-agility scanner. Every artifact is open at https://github.com/e2esolutions-tech/sezar under MIT.

Why QST specifically. The QKD axis treats channel-level quantum-secure key delivery as a measured property of an asset, not as a separate policy track. The emulator lets practitioners reproduce the channel-state classification work without QKD hardware. We hope this combination — a measurable QKD-aware posture model paired with a reproducible test rig — is useful to QST readers working on the deployment side of the field.

Submission disclosures

- The manuscript has not been published and is not under consideration at any other journal.
- Both authors approved the manuscript and the CRediT contributions stated in §10.
- We declare no competing interests. E2E Solutions develops and releases Sezar under MIT but derives no direct commercial revenue from the publication.
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- If accepted, we will reformat against the IOP iopart.cls template at the production stage.

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